2025 International Conference on Software, Knowledge, Information Management & Applications (SKIMA) | 978-1-6654-5734-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/SKIMA66621.2025.11155642

Machine Learning Approaches for Anomaly Detection in IoT Networks: A Survey

Kimsreng Lim
School of Graduate Studies
Cambodia University of Technology and Science
Phnom Penh, Cambodia
kl8024010014@camtech.edu.kh

Sophea Prum
School of Graduate Studies
Cambodia University of Technology and Science
Phnom Penh, Cambodia
sophea.prum@camtech.edu.kh

***Abstract*—The exponential growth of IoT devices across healthcare, smart cities, and industrial automation has heightened security challenges, rendering traditional detection systems inadequate for dynamic, heterogeneous networks. This survey provides a comprehensive review of machine learning techniques for anomaly detection in IoT networks, encompassing traditional methods, deep learning, and federated learning. We critically evaluate their effectiveness, focusing on IoT-specific challenges such as resource constraints, scalability, and concept drift. This survey also discusses recent advances such as lightweight machine learning models and privacy-preserving methods like federated learning, which show promise for improving deployment in IoT environments. The paper expands the dataset review with recent, underrepresented datasets tied to emerging IoT technologies and connects future research to trends like 5G integration and privacy preservation. By addressing practical trade-offs and outlining open research questions, this work offers actionable insights and a forward-looking roadmap for securing IoT systems.**

***Index Terms*—IoT Security, Anomaly Detection, Machine Learning, Deep Learning, Federated Learning, Intrusion Detection, IoT Networks**

I. Introduction

The rapid increase of Internet of Things (IoT) devices has transformed sectors such as healthcare, smart cities, and industrial automation (Fig. 1). However, this growth has amplified security risks due to the expanded attack surface and the dynamic, heterogeneous nature of IoT networks. Traditional signature-based detection systems often struggle in these environments, as they rely on predefined patterns that fail to capture emerging or evolving threats [1], [2]. Consequently, anomaly detection has emerged as a critical approach for identifying deviations from normal behavior, enabling the detection of previously unknown threats.

Hence, advanced machine learning (ML) techniques—ranging from decision trees to deep neural networks—have gained traction for anomaly detection in IoT. These techniques automatically learn patterns from network traffic and adapt to new attack types. They encompass traditional algorithms like decision trees and support vector machines (SVMs) as well as more sophisticated models such as convolutional neural networks (CNNs) and long short-term memory networks (LSTMs) [3], [4]. However, deploying these models in IoT networks involves trade-offs between accuracy, computational efficiency, and scalability, which are central to this survey.

This paper examines a variety of ML approaches for anomaly detection in IoT networks, including supervised, semi-supervised, and unsupervised learning models across traditional ML, deep learning, and federated learning paradigms. We also emphasize the importance of benchmarking using IoT-specific datasets [5], [6]. Our goal is to provide a comprehensive analysis of the state of the art, evaluating each approach's strengths and limitations while identifying potential future research directions and open challenges.

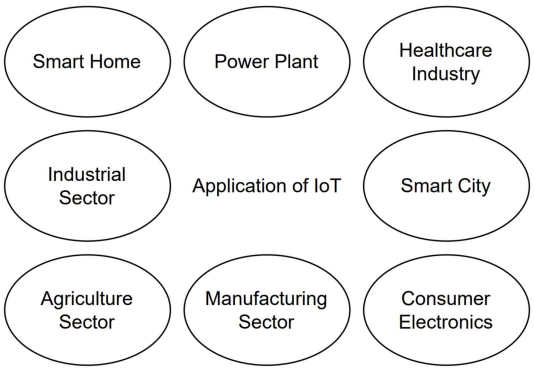


Fig. 1. Applications of IoT Devices Across Sectors

This paper is structured as follows: Section II provides an overview of IoT security challenges. Section III discusses edge computing and its role in IoT. Section IV presents a comparative analysis of traditional ML techniques. Section V provides an evaluation of a comparative framework for anomaly detection in IoT. Section VI explores deep learning approaches, while Section VII discusses federated learning method. Section VIII reviews key datasets for benchmarking. Finally, Section IX concludes with a road map for future research directions and open questions.

II. Overview of IoT Security Challenges

The IoT ecosystem comprises interconnected devices ranging from simple sensors to complex industrial systems. This diversity introduces significant security challenges. Resource constraints limit the computational power, memory, and energy

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on October 04,2026 at 12:27:53 UTC from IEEE Xplore. Restrictions apply.

of many IoT devices, making it difficult to deploy resource-intensive security solutions. Additionally, the heterogeneity of IoT networks—encompassing devices with different protocols, operating systems, and capabilities—complicates uniform security measures. Furthermore, IoT networks are highly dynamic, with devices frequently joining or leaving and threats evolving rapidly.

Traditional security approaches, such as signature-based intrusion detection systems (IDS), are unsuitable for IoT environments because they rely on known attack patterns and struggle to adapt to new threats [7], [8]. Anomaly detection, by contrast, identifies deviations from learned normal behavior, offering a more flexible and adaptive solution. However, anomaly detection in IoT faces challenges such as the scarcity of labeled data for supervised learning and the need for real-time detection in resource-constrained settings [5]. These issues have spurred the development of unsupervised and semi-supervised learning techniques, as well as federated learning approaches that enable decentralized, privacy-preserving model training [4].

Building on these security challenges and the need for real-time detection in constrained environments, the following section discusses edge computing—a promising strategy for distributing computational loads and enhancing privacy in IoT deployments.

III. Edge Computing

Edge computing has emerged as a key enabler for real-time data processing and low-latency applications in IoT networks. By processing data closer to the source—at the network edge—it reduces reliance on centralized cloud systems, enhancing responsiveness and privacy [9].

A. Architecture and Benefits

The edge computing architecture consists of three layers: the cloud, the edge, and IoT end devices. Application services, computing, and data storage are performed at the edge layer, minimizing latency and bandwidth usage by avoiding data transmission to remote cloud centers [10]. This setup is ideal for time-sensitive applications such as autonomous driving, healthcare monitoring, and smart cities [11].

B. Security Challenges

Despite its benefits, edge computing introduces new vulnerabilities. Edge devices, often deployed in untrusted environments, are susceptible to attacks like DDoS and malware injection [10]. Their limited computational resources also hinder robust security protocol implementation [11]. Solutions such as federated learning-based IDS and distributed firewalls are being explored [9], [12]. However, these solutions come with performance overhead—differential privacy may degrade accuracy, and homomorphic encryption is computationally expensive—underscoring the need for balanced security measures.

C. Privacy Considerations

Edge computing enhances privacy by keeping sensitive data local, but local data aggregation still poses risks. Techniques like homomorphic encryption and differential privacy are employed to protect data while maintaining functionality [10], [9]. Nevertheless, managing security across potentially thousands of edge nodes complicates operational overhead—especially when deploying frequent model updates or more advanced techniques like federated learning with encryption.

IV. Comparative Analysis of Conventional Machine Learning Techniques

Traditional ML techniques, such as decision trees, random forests, and SVMs, have been widely applied to anomaly detection in IoT networks due to their computational efficiency and interpretability, making them suitable for resource-constrained environments.

Ajeesh and Mathew [13] compared decision trees, random forests, and SVMs, finding that decision trees and random forests offer fast training and reasonable accuracy, while SVMs are more computationally expensive. Saied et al. [14] highlighted boosting algorithms like XGBoost and AdaBoost as effective for imbalanced datasets, though their computational demands may limit real-time IoT use. Table I shows results of conventional machine learning techniques from different studies.

TABLE I
Conventional Machine Learning Studies Comparison

Study	Model	Accuracy	Precision	Recall	F1-Score
[13]	RF	99.5%	99.2%	99.1%	99.1%
[14]	Gradient Boosting	99.3%	98.9%	99.0%	99.0%
[15]	RF	99.4%	99.1%	99.1%	99.1%
[16]	DT	99.0%	98.8%	98.7%	98.8%
[17]	AdaBoost	98.0%	97.5%	97.8%	97.6%

The choice of technique depends on the IoT context. Gradient Boosting offers high accuracy for imbalanced data but is computationally intensive, whereas decision trees suit constrained devices. Traditional methods struggle with high-dimensional data, motivating deep learning for capturing complex patterns in large datasets.

A. Learning Paradigm

Machine learning techniques for anomaly detection can be categorized into three primary paradigms: supervised, semi-supervised, and unsupervised learning. Supervised learning relies on labeled datasets to train models, making it highly effective but less adaptable to emerging threats [18], [19]. Semi-supervised learning, by contrast, leverages a combination of limited labeled data and large amounts of unlabeled data, balancing performance and adaptability [18]. Meanwhile, unsupervised learning methods detect anomalies without prior knowledge of normal behavior, making them well-suited for

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on October 04,2026 at 12:27:53 UTC from IEEE Xplore. Restrictions apply.

dynamic IoT environments, though they often suffer from high false positive rates [20].

B. Model Complexity and Computational Efficiency

Machine learning models also vary significantly in complexity and resource demands. Traditional models, such as decision trees, Naïve Bayes, and K-Means clustering, offer low computational cost and are suitable for resource-constrained IoT devices, though this comes at the expense of precision [19]. Deep learning-based approaches, including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and autoencoders, provide higher accuracy but require substantial computational resources [3]. Federated learning presents an alternative by enabling decentralized model training to preserve privacy, though it introduces communication overhead that can limit real-time deployment in IoT environments [12].

C. Deployment Suitability

Anomaly detection models can be deployed at different levels of the IoT architecture, each with unique advantages and limitations. On-device detection enables real-time anomaly identification but is constrained by hardware limitations. Edge-based detection offloads computation to intermediate edge nodes, balancing efficiency and processing power [10]. Cloud-based detection provides the highest computational capacity but introduces latency concerns and potential privacy issues, making it unsuitable for time-sensitive applications.

This taxonomy provides a structured perspective on IoT anomaly detection, aiding researchers in selecting appropriate machine learning techniques based on practical deployment constraints.

V. Comparative Framework for Anomaly Detection in IoT

While existing studies primarily compare machine learning models based on accuracy, real-world deployments require an assessment of additional factors such as computational cost, adaptability, and privacy considerations. Table II presents a comparative framework that evaluates different anomaly detection approaches based on these criteria.

TABLE II
Comparative Analysis of ML Techniques for IoT Anomaly Detection

Criterion	ML [4], [6]	DL [6], [12]	FL [8], [21]
Accuracy	Moderate	High	High
Computational Cost	Low	High	Medium
Training Data Needs	Moderate	High	High
Adaptability	Low	Medium	High
Privacy-Preserving	Low	Low	High
Real-Time Feasibility	High	Low	Medium

Lightweight machine learning models, such as decision trees and support vector machines (SVMs), provide moderate accuracy and low computational overhead, making them suitable for resource-limited IoT devices. In contrast, deep learning models, such as CNNs and LSTMs, offer higher accuracy but at the cost of significant computational and memory requirements. Federated learning techniques enable decentralized training, improving privacy preservation but introducing additional communication costs [1], [4]. This comparison provides insight into selecting the most appropriate method based on specific IoT deployment needs.

VI. Deep Learning for Anomaly Detection

Deep learning surpasses traditional techniques in capturing complex IoT anomalies. CNNs and LSTMs model high-dimensional patterns effectively, as shown by Butt et al. [1] and Gandi et al. [3]. However, deep models are resource-intensive, often exceeding IoT nodes' memory and processing limits. Balancing accuracy with efficiency is crucial, as *concept drift* demands frequent updates.

Beyond CNNs and LSTMs, autoencoders and generative models have also been employed to reconstruct normal IoT behaviors and detect deviations. Although these models can achieve high detection rates, their large parameter sizes often pose challenges in edge environments. Techniques such as model pruning, quantization, and knowledge distillation have been proposed to mitigate computational overhead, enabling near-real-time inference on resource-constrained devices. Moreover, adversarial robustness remains an open question; small perturbations to input signals may cause high misclassification rates, particularly for deep architectures lacking adversarial training.

Lightweight deep models [27] can partially address these issues by trading off some accuracy for lower memory footprints. Moreover, incremental and transfer learning methods allow models to adapt to evolving IoT traffic patterns while reducing the need for exhaustive retraining. Lastly, federated learning, discussed next, opens the door to continual edge training while preserving data privacy—especially important in sensitive domains such as healthcare and industrial IoT.

Table IV summarizes performance metrics from recent deep learning studies applied to IoT anomaly detection, highlighting the trade-offs between accuracy, precision, and computational complexity.

VII. Federated Learning in IoT Security

Federated learning (FL) enables decentralized model training, preserving privacy by keeping data on-device. Abbas et al. [12] achieved 99% accuracy using federated edge learning, demonstrating its potential for large-scale IoT networks. When combined with edge computing, FL reduces latency and bandwidth use, which is vital for applications like industrial automation. In practice, frameworks (e.g., PySyft or other open-source libraries) can help deploy FL with techniques like quantization or gradient sparsification to mitigate communication overhead. Fig. 2 illustrates the architecture of federated learning.

However, FL still faces challenges such as device heterogeneity and the high communication cost of model updates. Quantization and other compression strategies can alleviate

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on October 04,2026 at 12:27:53 UTC from IEEE Xplore. Restrictions apply.

TABLE III
EXPANDED IOT DATASETS COMPARISON

Dataset	Classes	Labeled	Environment	Data Types	Attack Types
TON IoT [22]	Multi-Class	Yes	Mixed IoT/IIoT	Telemetry, traffic, logs	Multiple
CICIDS2017 [23]	Multi-Class	Yes	Consumer IoT	Network traffic	Seven common attacks
CICIoT2023 [6]	Multi-Class	Yes	Consumer IoT	Network traffic	33 attacks
IoT-23 [17]	Multi-Class	Yes	Smart devices	Network traffic	Malware
BoT-IoT [24]	Multi-Class	Yes	Consumer IoT	Network traffic	DDoS, DoS, keylogging, data theft
N-BaIoT [21]	Multi-Class	Yes	Consumer IoT	Network telemetry	Mirai, BASHLITE, botnet attacks
Edge-IIoTset [25]	Multi-Class	Yes	Industrial IoT	Sensor data, system logs, network traffic	DDoS, MITM, injection, malware
SWaT [26]	Binary	Yes	Industrial IoT (ICS)	Sensor data, PLC traffic	Physical process manipulation

TABLE IV
DEEP LEARNING STUDIES COMPARISON

Study	Model	Accuracy	Precision	Recall	F1-Score
[1]	CNN + LSTM	94%	93%	94%	93.5%
[4]	DNN	91%	90%	91%	90.5%
[3]	CNN	97%	96%	97%	96.5%
[2]	VAE	93%	92%	93%	92.5%
[28]	EIDM	95%	94%	95%	94.5%

this overhead, but may reduce accuracy if not carefully managed.

FL's high accuracy in distributed settings underscores its promise for IoT security, though practical deployment requires careful consideration of communication overhead, heterogeneous devices, and ongoing security measures like adversarial defense.

As shown in Table V, federated learning studies demonstrate strong detection performance across various architectures, though performance varies with model type and communication optimization.

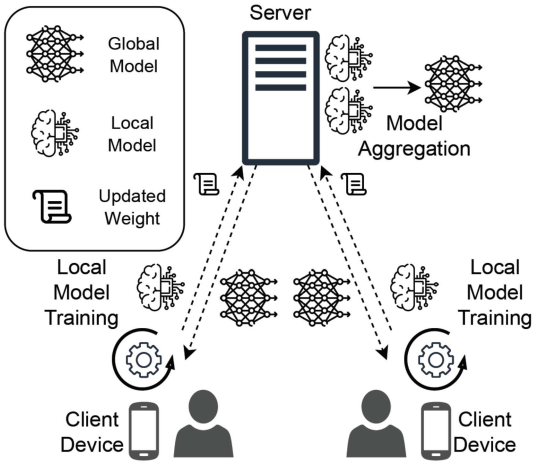


Fig. 2. Federated Learning Architecture

VIII. DATASETS FOR ANOMALY DETECTION IN IOT NETWORKS

High-quality datasets are essential for benchmarking anomaly detection models. This section reviews key datasets like TON_IoT, CICIDS2017, and CICIoT2023, as well as recent additions such as IoT-23, Edge-IIoTset, and MedBIoT, which reflect emerging trends in IoT security.

Table III provides a comparative analysis of these datasets, highlighting key attributes such as attack diversity, protocol coverage, and real-world applicability.

While these datasets provide valuable resources, gaps remain for 5G networks and large-scale heterogeneous environments. Collaborations with telecom providers or multi-institutional data-sharing frameworks could help fill these gaps, though privacy and commercial sensitivity remain hurdles. Additionally, synthetic dataset generators or real-time streaming IoT datasets could replicate large-scale traffic patterns when real data is limited.

TABLE V
FEDERATED LEARNING STUDIES COMPARISON

Study	Model	Accuracy	Precision	Recall	F1-Score
[12]	DNN	99.00%	99.00%	99.00%	99.00%
[4]	DNN	-	76.00%	61.00%	64.00%
[29]	GRU, LSTM	99.54%	98.94%	99.31%	98.60%

IX. CONCLUSION AND FUTURE DIRECTIONS

This survey consolidates machine learning approaches for anomaly detection in IoT networks, critically evaluating their strengths and limitations. Traditional ML techniques offer efficiency but struggle with complex patterns, while deep learning excels in accuracy at the cost of computational demands. Federated learning provides a promising path for privacy-preserving, decentralized detection, though communication overhead and device heterogeneity remain challenges.

Recent studies have investigated strategies such as combining lightweight models with privacy-preserving techniques like federated learning to balance efficiency, accuracy, and scalability. Based on trends and challenges identified in the reviewed literature, the following research directions are suggested to guide future work:

- **5G Integration:** As 5G networks roll out network slicing and ultra-low-latency demands, novel vulnerabilities and detection paradigms emerge, requiring models optimized for high-speed, dynamic environments.
- **Resource Constraints:** Exploring model compression techniques (e.g., pruning, quantization) can reduce overhead while maintaining performance in constrained IoT devices.

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on October 04,2026 at 12:27:53 UTC from IEEE Xplore. Restrictions apply.

- **Concept Drift and Continual Learning:** Frequent shifts in traffic patterns or device behaviors demand ongoing model updates, which federated learning can enable without centralized data storage.
- **Privacy Preservation:** Extending federated learning with lightweight encryption or differential privacy remains crucial for healthcare and other sensitive applications.

Researchers can leverage these insights by focusing on lightweight models for constrained devices and federated learning for privacy. A research roadmap includes short-term goals of optimizing lightweight models and compression for efficient edge inference. Medium-term efforts should reduce federated learning overhead through quantization and sparsification. In the long term, post-quantum security should be explored, alongside the development of standardized metrics to balance accuracy, resource efficiency, and privacy.

Open Research Questions

- **Standardized Trade-Off Measurements:** How can we effectively measure and compare the trade-off between model accuracy, resource usage, and privacy in a standardized way across varied IoT devices?
- **Handling Dynamic Data Streams:** What novel model update strategies can efficiently handle highly dynamic IoT data streams with minimal labeling requirements?
- **Adversarial Robustness:** How do we ensure the robustness of federated learning models against adversarial attacks in decentralized and heterogeneous environments?

By addressing these questions, the research community can further refine anomaly detection strategies, ensuring secure and resilient IoT networks for the foreseeable future.

REFERENCES

[1] N. Butt, A. Shahid, K. N. Qureshi, S. Haider, A. O. Ibrahim, F. Binzagr, and N. Arshad, "Intelligent deep learning for anomaly-based intrusion detection in IoT smart home networks," vol. 10, no. 23, p. 4598, publisher: MDPI AG. [Online]. Available: https://www.mdpi.com/2227-7390/10/23/4598

[2] A. Amin, A. Ganesh, and R. Piechocki, "A novel intrusion detection scheme using variational autoencoders," in *2023 International Symposium on Networks, Computers and Communications (ISNCC)*, pp. 1–7, ISSN: 2768-0940. [Online]. Available: https://ieeexplore.ieee.org/document/10323674

[3] V. P. Gandi, N. S. L. Jatla, G. Sadhineni, S. Geddamuri, G. K. Chaitanya, and A. K. Velmurugan, "A comparative study of AI algorithms for anomaly-based intrusion detection," in *2023 7th International Conference on Computing Methodologies and Communication (ICCMC)*, pp. 530–534. [Online]. Available: https://ieeexplore.ieee.org/document/10084186

[4] O. Belarbi, T. Spyridopoulos, E. Anthi, I. Mavromatis, P. Carnelli, and A. Khan, "Federated deep learning for intrusion detection in IoT networks." [Online]. Available: http://arxiv.org/abs/2306.02715

[5] J. White and P. Legg, "Unsupervised one-class learning for anomaly detection on home IoT network devices," in *2021 International Conference on Cyber Situational Awareness, Data Analytics and Assessment (CyberSA)*, pp. 1–8. [Online]. Available: https://ieeexplore.ieee.org/document/9478248

[6] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," vol. 23, no. 13, p. 5941. [Online]. Available: https://www.mdpi.com/1424-8220/23/13/5941

[7] M. F. Elrawy, A. I. Awad, and H. F. A. Hamed, "Intrusion detection systems for IoT-based smart environments: a survey," vol. 7, no. 1, p. 21. [Online]. Available: https://doi.org/10.1186/s13677-018-0123-6

[8] P. L. S. Jayalaxmi, R. Saha, G. Kumar, M. Conti, and T.-H. Kim, "Machine and deep learning solutions for intrusion detection and prevention in IoTs: A survey," vol. 10, pp. 121 173–121 192, conference Name: IEEE Access. [Online]. Available: https://ieeexplore.ieee.org/document/9941074

[9] K. Sha, T. A. Yang, W. Wei, and S. Davari, "A survey of edge computing-based designs for IoT security," vol. 6, no. 2, pp. 195–202. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S2352864818303018

[10] Y. Xiao, Y. Jia, C. Liu, X. Cheng, J. Yu, and W. Lv, "Edge computing security: State of the art and challenges," vol. 107, no. 8, pp. 1608–1631, conference Name: Proceedings of the IEEE. [Online]. Available: https://ieeexplore.ieee.org/document/8741060

[11] D. Xu, T. Li, Y. Li, X. Su, S. Tarkoma, T. Jiang, J. Crowcroft, and P. Hui, "Edge intelligence: Architectures, challenges, and applications." [Online]. Available: http://arxiv.org/abs/2003.12172

[12] S. Abbas, A. A. Hejaili, G. A. Sampedro, M. Abisado, A. S. Almadhor, T. Shahzad, and K. Ouahada, "A novel federated edge learning approach for detecting cyberattacks in IoT infrastructures," vol. 11, pp. 112 189–112 198.

[13] A. Ajeesh and T. Mathew, "Enhancing network security: A comparative analysis of deep learning and machine learning models for intrusion detection," in *2024 International Conference on E-mobility, Power Control and Smart Systems (ICEMPS)*, pp. 1–6. [Online]. Available: https://ieeexplore.ieee.org/document/10559350

[14] M. Saied, S. Guirguis, and M. Madbouly, "A comparative study of using boosting-based machine learning algorithms for IoT network intrusion detection," vol. 16, no. 1, p. 177. [Online]. Available: https://doi.org/10.1007/s44196-023-00355-x

[15] M. Hasan, M. M. Islam, M. I. I. Zarif, and M. M. A. Hashem, "Attack and anomaly detection in IoT sensors in IoT sites using machine learning approaches," vol. 7, p. 100059. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2542660519300241

[16] R. Vangipuram, R. K. Gunupudi, V. K. Puligadda, and J. Vinjamuri, "A machine learning approach for imputation and anomaly detection in IoT environment," vol. 37, no. 5, p. e12556. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1111/exsy.12556

[17] N. Sarwar, I. S. Bajwa, M. Z. Hussain, M. Ibrahim, and K. Saleem, "IoT network anomaly detection in smart homes using machine learning," vol. 11, pp. 119 462–119 480. [Online]. Available: https://ieeexplore.ieee.org/document/10287977

[18] G. Jayalaxmi and G. P. Babu, "Machine learning-based anomaly detection for securing iot networks: A review," *International Journal of Information Security*, vol. 21, no. 3, pp. 123–140, 2022.

[19] M. Saied and S. Ali, "Comparative analysis of machine learning techniques for anomaly detection in iot networks," *IEEE Internet of Things Journal*, vol. 10, no. 5, pp. 4301–4312, 2023.

[20] K. Sha and T. Yang, "A survey on edge computing for iot security: Challenges and solutions," *ACM Computing Surveys*, vol. 53, no. 3, pp. 1–35, 2020.

[21] Y. Meidan, M. Bohadana, Y. Mathov, Y. Mirsky, A. Shabtai, D. Breitenbacher, and Y. Elovici, "N-baiot—network-based detection of iot botnet attacks using deep autoencoders," *IEEE Pervasive Computing*, vol. 17, no. 3, pp. 12–22, 2018.

[22] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "TON_iot telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems," vol. 8, pp. 165 130–165 150. [Online]. Available: https://ieeexplore.ieee.org/document/9189760/

[23] M. Catillo, A. Del Vecchio, A. Pecchia, and U. Villano, "A case study with CICIDS2017 on the robustness of machine learning against adversarial attacks in intrusion detection," in *Proceedings of the 18th International Conference on Availability, Reliability and Security*, ser. ARES '23. Association for Computing Machinery, pp. 1–8. [Online]. Available: https://doi.org/10.1145/3600160.3605031

[24] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, "Towards the Development of Realistic Botnet Dataset in the Internet of Things for Network Forensic Analytics: Bot-IoT Dataset," Nov. 2018, arXiv:1811.00701 [cs]. [Online]. Available: http://arxiv.org/abs/1811.00701

[25] M. A. Ferrag, O. Friha, D. Hamouda, L. Maglaras, and H. Janicke, "Edge-iiotset: A new comprehensive realistic cyber security dataset of iot and iiot applications for centralized and federated learning," *IEEE Access*, vol. 10, pp. 40 281–40 306, 2022.

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on October 04,2026 at 12:27:53 UTC from IEEE Xplore. Restrictions apply.

[26] J. Goh, S. Adepu, K. Junejo, and A. Mathur, "A dataset to support research in the design of secure water treatment systems," 10 2016.
[27] M. M. Khan and M. Alkhathami, "Anomaly detection in IoT-based healthcare: machine learning for enhanced security," vol. 14, no. 1, p. 5872, publisher: Nature Publishing Group. [Online]. Available: https://www.nature.com/articles/s41598-024-56126-x
[28] O. Elnakib, E. Shaaban, M. Mahmoud, and K. Emara, "EIDM: deep learning model for IoT intrusion detection systems," vol. 79, no. 12, pp. 13 241–13 261. [Online]. Available: https://doi.org/10.1007/s11227-023-05197-0
[29] V. Mothukuri, P. Khare, R. M. Parizi, S. Pouriyeh, A. Dehghantanha, and G. Srivastava, "Federated-learning-based anomaly detection for IoT security attacks," vol. 9, no. 4, pp. 2545–2554. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9424138

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on October 04,2026 at 12:27:53 UTC from IEEE Xplore. Restrictions apply.